

# 1 Non-dimensionalisation

in terms of  $c = \sqrt{gH}$ , gravitational wave speed and  $Fr = \frac{U}{c}$ , Froude number

$$\frac{\partial u}{\partial t} + c \cdot Fr \frac{\partial u}{\partial x} + c \frac{\partial h}{\partial x} = 0$$

$$\frac{\partial h}{\partial t} + c \cdot Fr \frac{\partial h}{\partial x} + c \frac{\partial u}{\partial x} = 0$$

which can be written as system

$$\frac{\partial}{\partial t} \begin{pmatrix} u \\ h \end{pmatrix} = - \underbrace{c \cdot Fr \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \frac{\partial}{\partial x} \begin{pmatrix} u \\ h \end{pmatrix}}_{A_u} - \underbrace{c \cdot Fr \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \frac{\partial}{\partial x} \begin{pmatrix} u \\ h \end{pmatrix}}_{A_h} - \underbrace{c \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \frac{\partial}{\partial x} \begin{pmatrix} u \\ h \end{pmatrix}}_F - \underbrace{c \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \frac{\partial}{\partial x} \begin{pmatrix} u \\ h \end{pmatrix}}_G$$

## 1.1 Semi-implicit

$$\begin{aligned} u' &= u^{(n)} - \Delta t c \cdot Fr \left( (1 - \alpha) u_x^{(n)} + \alpha u_x^{(i-1)} \right) - \Delta t c (1 - \alpha) h_x^{(n)} \\ h^i &= h^{(n)} - \Delta t c \cdot Fr \left( (1 - \alpha) h_x^{(n)} + \alpha h_x^{(i-1)} \right) - \Delta t c \left( (1 - \alpha) u_x^{(n)} + \alpha (u'_x - \Delta t \alpha h_{xx}^i) \right) \\ u^i &= u' - \Delta t \alpha h_{xx}^i \end{aligned}$$

## 1.2 Segregated

$$\begin{aligned} u' &= u^{(n)} - \Delta t c \cdot Fr \left( (1 - \alpha) u_x^{(n)} + \alpha u'_x \right) - \Delta t c \left( (1 - \alpha) h_x^{(n)} + \alpha h_x^{i-1} \right) \\ u'' &= u^{(n)} - \Delta t c \cdot Fr \left( (1 - \alpha) u_x^{(n)} + \alpha u'_x \right) - \Delta t c \left( (1 - \alpha) h_x^{(n)} \right) \\ h^i &= h^{(n)} - \Delta t c \cdot Fr \left( (1 - \alpha) h_x^{(n)} + \alpha h_x^{i/(i-1)} \right) - \Delta t c \left( (1 - \alpha) u_x^{(n)} + \alpha (u''_x - \Delta t \alpha h_{xx}^i) \right) \\ u^i &= u'' - \Delta t \alpha h_{xx}^i \end{aligned}$$

## 1.3 Operator Split

$$\begin{aligned} u' &= u^{(n)} - \Delta t c \cdot Fr \left( (1 - \alpha) u_x^{(n)} + \alpha u'_x \right) \\ h^i &= h^{(n)} - \Delta t c \cdot Fr \left( (1 - \alpha) h_x^{(n)} + \alpha h_x^{i/(i-1)} \right) - \Delta t c \left( (1 - \alpha) \left( u_x^{(n)} - \Delta t c (1 - \alpha) h_{xx}^{(n)} \right) + \alpha (u''_x - \Delta t c \alpha h_{xx}^i) \right) \\ u^i &= u'' - \Delta t (1 - \alpha) h_{xx}^{(n)} - \Delta t \alpha h_{xx}^i \end{aligned}$$

## 2 Single wave-mode solution

$$\begin{pmatrix} u \\ h \end{pmatrix} = \begin{pmatrix} u_0 \\ h_0 \end{pmatrix} e^{i(kx - \omega t)}$$

$$\implies \frac{u_0}{h_0} = \pm 1$$

$$\omega = kc(Fr \pm 1)$$

for which the operators map

$$A_u \mapsto -ikcFr \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$A_h \mapsto -ikcFr \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$F \mapsto -ikc \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$G \mapsto -ikc \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

## 3 Numerical dispersion

for a centred-difference scheme (where  $j$  denotes the spatial grid point)

$$\frac{\partial}{\partial x} y_j \rightarrow \frac{y_{j+1} - y_{j-1}}{2\Delta x}$$

for a wave mode,  $y_j = e^{ikx_j}$  for any  $j$  so

$$\begin{aligned} \frac{\partial}{\partial x} y_j &\rightarrow \frac{1}{2\Delta x} (e^{ikx_j + \Delta x} - e^{ikx_j - \Delta x}) \\ &\rightarrow \frac{1}{2\Delta x} (e^{ik\Delta x} - e^{-ik\Delta x}) e^{ikx_j} \\ &\rightarrow \frac{i}{\Delta x} \sin(k\Delta x) e^{ikx_j} \end{aligned}$$

so, instead of the operators in section 2, the following should be used:

$$A_u \mapsto -\frac{i}{\Delta x} \sin(k\Delta x) cFr \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$A_h \mapsto -\frac{i}{\Delta x} \sin(k\Delta x) c Fr \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$F \mapsto -\frac{i}{\Delta x} \sin(k\Delta x) c \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$G \mapsto -\frac{i}{\Delta x} \sin(k\Delta x) c \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

## 4 Formulating in terms of Courant numbers (spatial discretisation-independent)

Given the update for any iteration of a numerical scheme for the linear SWE can be written in the form  $\begin{pmatrix} u^i \\ h^i \end{pmatrix} = M_1 \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix} + M_2 \begin{pmatrix} u^{i-1} \\ h^{i-1} \end{pmatrix} = M \begin{pmatrix} u^{(n)} \\ h^{(n)} \end{pmatrix}$ , the scheme is stable where the eigenvalues of matrix  $M$ ,  $|\lambda_\alpha| \leq 1$ , so the problem of stability reduces to finding the region where this holds, in terms of the advective Courant number,  $c_a$ , and the gravitational Courant number,  $c_g$ .

$$\begin{aligned} c_a &= kcFr\Delta t \\ c_g &= kc\Delta t \\ \implies Fr &= \frac{c_a}{c_g} \text{ and } c = \frac{ic_g}{k\Delta t} \end{aligned}$$

With these substitutions, the matrix  $M$  for any iteration of each numerical scheme can be written in terms of the variables  $c_a$ ,  $c_g$ . Determining the magnitude of the eigenvalues across a range of values will indicate whether or not the scheme is stable for these values.